

TRANSPORT AND THE SUPERCRITICALITY GAP IN 3D NAVIER–STOKES, III: THE LEDGER

ADILKHAN SALKIMBAYEV

ABSTRACT. Part II reduced the regularity problem for the three-dimensional Navier–Stokes equations to control of the stochastic flow map, and showed that any such control must use the coupling in which the flow carries and stretches the vorticity and the Biot–Savart law turns the carried vorticity back into velocity. Here we write that coupling as a ledger. Combining the stochastic Cauchy formula of Constantin and Iyer with the Biot–Savart law, the strain felt by a fluid particle is a singular integral, over an independent copy of the stochastic flow, of the stretched initial vorticity of all other labels, weighted at their current separation; the viscous equations are, in these variables, a mean-field system. We give an exact Fourier characterization of the part of the strain supplied by sources in a distance shell, which makes the ledger computable. We list the known blowup and growth constructions for neighbouring equations by the property of the ledger each one alters. We then report two registered numerical campaigns. The first measures, per strain-producing event, how production is handed to the next smaller scale: events ending in roll-up forward production, events ending in pressure-driven flattening mostly do not, no production escalates at the critical rate, and successive escalations are weakly positively correlated in one of two flows. The second attributes the strain eigenvalues of producing events to their sources: sources at one half to two Taylor lengths pay most of the compression, the share of the event’s own structure grows with intensity, and structures support rather than attenuate their own sheet formation. We write the escalation between events as an exact bookkeeping identity for the event Reynolds number, which makes the critical escalation rate precise and states the missing estimate as a question about chains of events.

1. INTRODUCTION

Part I [16] eliminated, term by term, every term on the right-hand side of the strain equation as the carrier of a critical estimate, and showed that advection is invisible to every Lebesgue budget. Part II [17] moved to the flow map and proved three things: regularity on a finite interval is equivalent to a bound on the gradient of the stochastic flow; the geometry that the energy controls cannot see deformation; and the a priori Lagrangian bound obtained from the dissipation is exactly as supercritical as the energy. The target became a statement on the concentration of the stochastic flow, which must use the Cauchy–Weber coupling [17, §6].

The known blowup constructions for neighbouring equations each describe that coupling precisely, but only for one scenario. None of them states the general principle by which the coupling escalates or fails to. This paper does three things towards it. Section 2 writes the coupling as a ledger in Lagrangian variables and gives an exact tool for splitting it by source distance. Section 3 records, for each known construction, which property of the ledger it alters: these are constraints that any general principle must satisfy, not the principle itself. Sections 4 and 5 report two registered numerical campaigns that measure the transactions in real flows.

What is new, and what is not. The stochastic Lagrangian representation and its mean-field (“nonlinear in the sense of McKean”) character are due to Constantin and Iyer [4]; the Lagrangian

form of the Biot–Savart law is standard in vortex methods. The Eulerian decomposition of strain into local and nonlocal parts is due to Hamlington, Schumacher and Dahm [7] and was used by Buaria, Pumir and Bodenschatz [1, 2]. What we add: the use of the Lagrangian Biot–Savart form as an accounting ledger with producing events as units (Proposition 2.1); the exact Fourier multiplier of a distance shell (Lemma 2.2), which we have not seen used for this purpose; the constraints table; and the two registered campaigns, conditioned per event and per exit channel.

2. THE LEDGER

Let u be a smooth solution of the Navier–Stokes equations on $\mathbb{T}^3$ on $[0, T^*)$, with viscosity ν , vorticity ω and strain S . The strain is recovered from the vorticity by the Biot–Savart law,

$$S(x) = \text{p.v.} \int K_S(x - y) \omega(y) dy, \quad K_S(z)_{ij,l} = -\frac{3}{8\pi} \frac{(\epsilon_{ilm} z_j + \epsilon_{jlm} z_i) z_m}{|z|^5}, \quad (1)$$

on $\mathbb{R}^3$; on $\mathbb{T}^3$, K_S denotes the kernel of the Fourier multiplier $\hat{\omega} \mapsto \hat{S}$ on mean-zero fields. The symmetric part of ∇u has no local (delta) term: the local term of ∇u is antisymmetric.

Let X be the stochastic flow of Constantin and Iyer [4], $dX_t = u(X_t, t) dt + \sqrt{2\nu} dW_t$, $X_0 = \text{id}$, with $F = \nabla X$ and $A = X^{-1}$, and let X' , F' , A' be an independent copy.

Proposition 2.1 (the ledger). *For $t < T^*$ and $x \in \mathbb{T}^3$,*

$$S(x, t) = \mathbb{E}' \text{p.v.} \int_{\mathbb{T}^3} K_S(x - X'_t(b)) F'_t(b) \omega_0(b) db, \quad (2)$$

where the principal value is taken over preimages under X'_t of balls centred at x . For $\nu = 0$ the expectation is absent and (2) is the classical Lagrangian form of the Biot–Savart law.

Proof. By the stochastic Cauchy formula [4, Prop. 2.7], $\omega(y, t) = \mathbb{E}'[(F'_t \omega_0)(A'_t(y))]$. Insert this into (1), exchange the expectation and the integral (everything is smooth and bounded for $t < T^*$), and in each realization change variables $y = X'_t(b)$, which preserves volume. $\square$

By [17, Lemma 2.2], a material line element $F_t(a)\xi$ carried by the particle's own path stretches at the rate $\hat{e} \cdot S(X_t(a), t) \hat{e}$. So the stretching of one particle is paid for by all other labels b : each contributes its initial vorticity $\omega_0(b)$, stretched by its own history $F'_t(b)$, weighted by the kernel at its current separation $X_t(a) - X'_t(b)$ from the particle, and averaged over an independent copy of the noise. This is the object that single-time Lebesgue budgets cannot see [16, Prop. 5.1]: a norm of $\omega(t)$ records the stretched vorticity at each point, but not which label carries it, how much that label has been stretched, or with which particle it is paired. In the inviscid case the pairing is deterministic; with viscosity the particle is paired with an average, which is where a decorrelation of successive events, if there is one, must come from. It cannot come from the noise acting on pairs of labels directly: the noise does not depend on the label, so it cancels in the separation $X_t(a) - X_t(b)$, and Mahithitarmmatorn [9] shows that the two-point operator of the stochastic Weber field is degenerate in the separation variable. Any decorrelation must come through the average that defines the field, that is, through the self-consistency of (2).

Splitting the ledger by source distance. To attribute the strain to its sources we restrict the kernel in (1) to a shell of source distances. For $0 \leq a < b \leq \infty$ let $S^{[a,b]}$ be the strain induced by the vorticity at distances $a \leq |x - y| < b$ (on $\mathbb{T}^3$ with the minimal-image distance, for $b \leq \pi$; the outer shell is defined by complement).

Lemma 2.2 (shell multiplier). $\widehat{S^{[a,b]}}(k) = g(a, b, |k|) \hat{S}(k)$, where

$$g(a, b, \kappa) = \frac{h(b\kappa) - h(a\kappa)}{h(\infty)}, \quad h(x) = \int_0^x \frac{j_2(s)}{s} ds, \quad h(\infty) = \frac{1}{3},$$

and j_2 is the spherical Bessel function of order two.

Proof. The symbol $\hat{S}(k) = -\frac{1}{2}[\hat{k} \otimes (\hat{k} \times \hat{\omega}) + (\hat{k} \times \hat{\omega}) \otimes \hat{k}]$ is homogeneous of degree zero and quadratic in $\hat{k}$, so each component is a combination of spherical harmonics of degrees 0 and 2; the degree-zero part is the spherical mean, proportional to $\epsilon_{ijm} + \epsilon_{jim} = 0$. The kernel of a degree-zero multiplier whose symbol is a degree-two harmonic $Y(\hat{k})$ is $cY(\hat{z})|z|^{-3}$. By the Funk–Hecke formula, the Fourier transform of $Y(\hat{z})|z|^{-3}\mathbf{1}_{a \leq |z| < b}$ is $-4\pi Y(\hat{k}) \int_a^b j_2(r|k|) r^{-1} dr = -4\pi Y(\hat{k})(h(b|k|) - h(a|k|))$, and letting $a \rightarrow 0$, $b \rightarrow \infty$ identifies the full symbol. On $\mathbb{T}^3$ a kernel supported in $|z| < \pi$ has Fourier coefficients equal to its transform on $\mathbb{R}^3$ at integer k . Finally $h(\infty) = \sqrt{\pi/2} \int_0^\infty J_{5/2}(s) s^{-3/2} ds = \sqrt{\pi/2} \Gamma(1)/(2^{3/2} \Gamma(5/2)) = 1/3$ by the Weber–Schafheitlin integral. $\square$

Numerically, the shells of Lemma 2.2 sum to the total strain to 4×10^{-16} , and a direct physical-space quadrature of (1) over the shell $[0.5, 1.5)$ agrees with the multiplier to 2.7×10^{-2} at 64^3 and 3.2×10^{-3} at 128^3 .

3. CONSTRAINTS FROM THE KNOWN CONSTRUCTIONS

Navier–Stokes on $\mathbb{T}^3$ runs the ledger (2) with five properties at once: (K) the exact, translation-invariant kernel K_S ; (C) smooth data and solutions; (W) the Cauchy–Weber form, in which the stretched vorticity is carried by the flow; (D) transport by the flow itself; (N) the noise average $\mathbb{E}'$, that is, viscosity; and no boundary. Every known blowup or small-scale-growth construction for a neighbouring equation alters at least one of these. Table 1 lists them. A general principle deciding escalation must fail in each row and hold for Navier–Stokes; the rows are constraints on it, not instances of it.

Two observations. First, the two constructions that keep viscosity alter the nonlinearity itself: (W) or (D). No known construction keeps (K), (C), (W), (D) and (N) together. Second, the claimed Euler construction of [13], which we have read only in part and not verified, is organized as an escalation chain in the sense of Part I: a sequence of localized packets in which the gradient of each packet becomes the shear that amplifies the next, with a gain per stage and stage durations whose sum is finite. In ledger terms it is a chain in which each event’s output is the next event’s stretcher and the escalation compounds perfectly. By its own scales it is structurally inviscid: the viscous decay of each packet over its stage vastly exceeds its gain. If the construction is correct, the row it occupies is exactly the one that the noise average (N) must defeat.

4. THE HAND-OFF (CAMPAIGN II-A)

Setup. Pseudo-spectral simulations on $\mathbb{T}^3$ at 128^3 , $\nu = 1/400$, $0 \leq t \leq 10$, from the Taylor–Green vortex (**tg**) and a random-phase field (**rand**), both of energy $1/8$; Taylor-scale Reynolds number about 40 at peak dissipation [16, §6]. 32^3 tracers, advanced with the velocity, sample every 0.05 the Gaussian-filtered gradients $\nabla(G_j * u)$, $\hat{G}_j(k) = \exp(-|k|^2/2k_j^2)$, for $k_j \in \{4, 8, 16\}$. At each filter scale a visit to the producing set $\Sigma_j = \{-\det S_j > 0, \operatorname{tr}(\nabla u_j)^2 > 0\}$ ends by roll-up or by flattening [16, §6.5]. For a visit at scale j ending at t_e , over $W = [t_e, t_e + 0.5]$:

- the *enhancement* E_c of production at scale $j+1$ after exits by channel c , relative to a time-matched tracer average, weighted by the production of the visit;
- the *escalation exponent* $\alpha = \log_2(\max_W |S_{j+1}| / \max_V |S_j|)$, against the critical value $\alpha = 2$ ($|\nabla u| \propto \ell^{-2}$) and the typical value $\alpha_0 = \log_2(\operatorname{rms}|S_{j+1}| / \operatorname{rms}|S_j|)$;
- for chains (a scale-4 visit followed within W by a scale-8 visit of the same tracer) the Spearman correlation $\rho(\alpha_1, \alpha_2)$ of successive escalations.

construction	result	property of the ledger altered
Kiselev–Šverák [8]	double-exponential gradient growth, 2D Euler on a disk	boundary (image kernel); 2D, so the Cauchy formula carries ω_0 without stretching; (N) absent
Elgindi [5]	finite-time blowup, $C^{1,\alpha}$ 3D Euler on $\mathbb{R}^3$	(C): at leading order in small α the Biot–Savart law reduces to a simpler, essentially local operator; (N) absent
Chen–Hou [3]	blowup, smooth 3D axisymmetric Euler with boundary	boundary (image kernel); (N) absent
Tao [15]	blowup, Euler with modified inertia operator, on $\mathbb{R}^3$	(K): kernel not translation-invariant (inhomogeneous, non-isotropic, within ε of Euler’s metric); retains (W), (D), Kelvin, helicity; (N) absent
Tao [14]	blowup, averaged Navier–Stokes	(W): nonlinearity is an average, not the transport of stretched vorticity; (N) present
Miller [10]	blowup, strain self-amplification model	(D): no transport; (N) present
OpenAI [13], unverified	claimed blowup, smooth unforced 3D Euler on $\mathbb{R}^3$	(N) absent only, if correct

TABLE 1. Known constructions and the property of the ledger each alters.

Registered pictures: (II.1) $E_{\text{roll}} > E_{\text{flat}}$ at both steps in both runs; (II.2) production at $\alpha \geq 2$ below 1%, and below 5% in the top 1% of visits; (II.3) $\rho \leq 0.1$ in both runs. An instrument gate required a surrogate enhancement, from random tracer-times, in $[0.9, 1.1]$. A 64^3 smoke test failed the gate with a global baseline (1.118); the time-matched baseline above was adopted before the campaign, with hypotheses unchanged. The smoke-test values were seen before the campaign, which is therefore a replication at higher resolution, not a blind test.

Results. The gate passed in all eight checks (0.94–1.06).

	tg growth	tg resolved	rand growth	rand resolved
$E_{\text{roll}}, 4 \rightarrow 8 / 8 \rightarrow 16$	1.09/2.25	1.13/2.54	1.44/1.37	1.56/1.48
$E_{\text{flat}}, 4 \rightarrow 8 / 8 \rightarrow 16$	0.51/1.52	−0.11/0.82	0.44/0.14	0.34/0.11
production at $\alpha \geq 2$	$< 5 \cdot 10^{-5}$	$< 5 \cdot 10^{-5}$	$< 5 \cdot 10^{-5}$	$< 5 \cdot 10^{-5}$
max $\alpha, 4 \rightarrow 8 / 8 \rightarrow 16$	3.3/3.9	3.3/3.9	3.0/2.0	2.0/1.7
mean α (α_0), $4 \rightarrow 8$	−0.21 (0.43)	−0.02 (0.28)	0.37 (0.74)	0.45 (0.67)
chains; $\rho(\alpha_1, \alpha_2)$	3453; +0.24	399; −0.16	352; −0.09	53; +0.56

TABLE 2. Campaign II-A. Growth: $t \leq 6.75$ (tg), $t \leq 2.25$ (rand); resolved: $t \leq 4.5$, $t \leq 1.25$.

II.1 and II.2 passed; II.3 failed as registered. Events ending in roll-up hand production to the next smaller scale; events ending in pressure-driven flattening are mostly followed by below-background production at the next scale. No production escalates at or above the critical

rate, although isolated hand-offs reach $\alpha = 3\text{--}4$; on average intense events escalate less than typical ones. Successive escalations are positively correlated in the turbulent phase of **tg** ($\rho = 0.24$ over 3453 chains, standard error about 0.017) and not in **rand**. The two exponents share a peak value of $|S_8|$ with opposite signs, which biases ρ downward, so the **tg** value is if anything an underestimate. The resolved windows disagree with each other, one on only 53 chains. The picture that survives is not “successive events are independent” but “escalation compounds along the roll-up channel while staying well below the critical rate”.

5. THE TRANSACTIONS (CAMPAIGN II-B)

Setup. The same simulations and tracers. At each sample the strain at every tracer is split by Lemma 2.2 into five source shells, $[0, 0.1)$, $[0.1, 0.2)$, $[0.2, 0.4)$, $[0.4, 0.8)$, $[0.8, \infty)$, grouped as SELF (< 0.2 , up to about six Kolmogorov lengths), PARENT ($0.2\text{--}0.8$, about one half to two Taylor lengths, the range of the stretchers of [6]) and BACKGROUND. With e_i the eigenvectors of the total strain, $a_i^{(s)} = e_i \cdot S^{(s)} e_i$ sums over shells to λ_i exactly. Registered hypotheses: (T1) PARENT and BACKGROUND supply more than half of λ_1 at producing samples; (T2) SELF opposes sheet formation, $\sum_P a_2^{\text{SELF}} \leq 0$; (T3) the PARENT share of λ_1 is larger before roll-up exits than before flattening exits. Kill: T1 failing in both runs. The registration stated $h(\infty) = 2/3$; the correct value is $1/3$ (Lemma 2.2), confirmed by the direct quadrature; this was an arithmetic slip in writing the registration, and the hypotheses are unchanged.

Results. The shells summed to λ_1 to single precision (2×10^{-7}). Table 3 gives the production-weighted shares SELF / PARENT / BACKGROUND.

	tg growth	tg resolved	rand growth	rand resolved
producing, λ_1	.28/.58/.14	.20/.53/.26	.30/.62/.08	.21/.67/.12
producing, λ_2	.28/.48/.24	.19/.51/.31	.26/.63/.12	.21/.67/.12
top 1%, λ_1	.35/.58/.07	.30/.57/.14	.41/.53/.05	.31/.61/.08
before roll-up, λ_1	.39/.53/.09	.30/.56/.15	.35/.58/.07	.23/.66/.11
before flattening, λ_1	.43/.51/.06	.25/.58/.18	.41/.54/.04	.21/.69/.10
after roll-up, λ_1	.23/.69/.09	.21/.69/.11	.35/.58/.06	.26/.65/.09
after flattening, λ_1	.19/.73/.08	.25/.67/.09	.32/.61/.07	.19/.72/.10

TABLE 3. Campaign II-B: who pays for the strain of producing events. “Before” is the last 0.25 of a visit; “after” is the producing samples in the 0.5 following an exit.

T1 passed in both runs and both windows: sources beyond about six Kolmogorov lengths pay 70–80% of the compression. T2 failed in both runs: the SELF shells *support* sheet formation, supplying 19–28% of λ_2 . This does not contradict the self-attenuation of [1], which is conditioned on extreme vorticity rather than on producing strain. T3 passed as registered in the growth windows, by margins of 0.015 and 0.035, and reversed in both resolved windows; we read it as no reliable dependence of the payer on the exit channel. Three findings have the same sign in all four windows. PARENT, the sources at one half to two Taylor lengths, is the principal payer, with 48–73% of both eigenvalues. As an event intensifies (all producing samples against the top 1%) the SELF share rises, from .28, .20, .30, .21 to .35, .30, .41, .31, and the BACKGROUND share falls, from .14, .26, .08, .12 to .07, .14, .05, .08, while PARENT stays roughly constant. Intense events are more self-supported and less driven by the large scales.

Combined with §4: events that end in roll-up and events that end in flattening are paid for in the same way. The exit channel decides *whether* an event at the next smaller scale follows, not who pays for it.

6. THE ESCALATION RECURSION

A *hand-off chain* is a sequence of producing events $E_0, E_1, \dots$, not necessarily along one trajectory, with scales $\ell_0 > \ell_1 > \dots$ and peak strain intensities I_k . Define the escalation exponent and the event Reynolds number by

$$\alpha_k = \frac{\log(I_{k+1}/I_k)}{\log(\ell_k/\ell_{k+1})}, \quad R_k = \frac{I_k \ell_k^2}{\nu}.$$

R_k is the ratio of the viscous time ℓ_k^2/ν to the strain time $1/I_k$; an event with R_k below a threshold R_c is dissipated before it produces.

Lemma 6.1 (bookkeeping). $\log(R_n/R_0) = \sum_{k < n} (\alpha_k - 2) w_k$, with $w_k = \log(\ell_k/\ell_{k+1}) > 0$.

Proof. $R_{k+1}/R_k = (I_{k+1}/I_k)(\ell_{k+1}/\ell_k)^2$; take logarithms. $\square$

The value $\alpha = 2$ is the scaling of the Navier–Stokes equations: R is invariant exactly when $I \propto \ell^{-2}$.

Corollary 6.2. *If a chain has $\ell_n \rightarrow 0$ and $R_n \geq R_c$ for all n , then the w -weighted mean of $\alpha_0, \dots, \alpha_{n-1}$ is at least $2 - \log(R_0/R_c)/\log(\ell_0/\ell_n)$, which tends to 2.*

This is exact but conditional in its use. Consider the *event picture*. Growth of $\|S\|_{L^\infty}$ proceeds through producing events. Each event lasts a bounded number of strain times and changes $\log \|\nabla X\|$ by a bounded amount ([17, Cor. 2.3]; Table 4 of [16]). Under the event picture, a singularity needs infinitely many events in finite time, reaching arbitrarily small scales above the viscous threshold, and so, by Corollary 6.2, chains whose mean escalation reaches the critical value 2.

Question 6.3 (event-level criticality). *Is there $\delta > 0$ such that, for the Navier–Stokes equations, every hand-off chain with $R_k \geq R_c$ has w -weighted mean escalation at most $2 - \delta$?*

A positive answer, together with the event picture, excludes singularities. It is the event-level counterpart of the Lagrangian question [17, Question 6.1], and like it cannot follow from Lebesgue budgets [16, Prop. 5.1] or from Lagrangian bookkeeping driven by the dissipation alone [17, Prop. 5.4].

What the measurements say, and what they cannot say. In the simulations the production-weighted mean of α is between -0.23 and 0.37 at the two steps, with standard deviation 0.37 – 0.75 (an exploratory computation from the data of §4, not registered). By count, 0.3% (**tg**) and 0.08% (**rand**) of hand-offs reach $\alpha \geq 2$, and they carry almost no production. Successive exponents are correlated at most weakly ($\rho \leq 0.24$). If successive α_k formed a stationary Gaussian sequence with lag-one correlation ρ , the probability that the mean over n steps reaches 2 would decay like $\exp\{-n(2 - \mu)^2(1 - \rho)/(2\sigma^2(1 + \rho))\}$. Compounding enlarges the effective variance by $(1 + \rho)/(1 - \rho) \approx 1.6$ at $\rho = 0.24$ but leaves the drift $\mu - 2$, between -1.6 and -2.2 , unchanged; with the measured μ and σ and $\rho = 0.24$ the rate is between about 2.6 and 9 per step. Typical chains are therefore strongly subcritical. This says nothing about the worst case, which is what Question 6.3 asks about. A singularity, if one exists, arises from special data, not typical data.

The ledger identifies who sets α . PARENT-distance sources pay most of each event's compression, SELF pays a share that grows with intensity, and the large scales recede. A chain can therefore run away in two ways, and Table 1 and the literature constrain both.

- *Self-sustained*: the SELF share tends to one as the intensity grows, so that the structure drives its own collapse. This is the self-similar ansatz, excluded in the natural classes by Nečas, Růžička and Šverák [11] and by Tsai [12]. In the simulations the SELF share rises with intensity but stays at or below 0.41 in the top 1%.
- *Parent-driven*: each event's output drives the next at $\alpha \geq 2$. This is the shape of Tao's averaged construction [14], which gives up the Cauchy–Weber form, and of the claimed Euler construction [13], which gives up viscosity.

The missing principle, in this language, is the property of the exact averaged ledger (2) that keeps the parent-driven escalation below $2 - \delta$ when both the Cauchy–Weber form and the noise average are present. We do not have it. Part IV [18] replaces the event picture by rigorous definitions, identifies circulation as the currency of the ledger, and proves, in an exact reduction, the viscous threshold that the event picture assumes.

ACKNOWLEDGEMENTS

The direction of this work, including the reading of the transport as an accounting problem and the insistence on a general principle rather than scenario-specific formulas, originates with the author. The derivations, the numerical campaigns, the literature search and a first draft of this text were produced in collaboration with Claude, a generative AI system developed by Anthropic, working under the author's direction. The author is responsible for the contents of this paper, including any errors. Every numerical hypothesis was registered, with its kill criterion, before the run that tested it; amendments made before the runs are stated in the text.

REFERENCES

- [1] D. Buaria, A. Pumir and E. Bodenschatz, *Self-attenuation of extreme events in Navier–Stokes turbulence*, Nat. Commun. **11** (2020), 5852.
- [2] D. Buaria, E. Bodenschatz and A. Pumir, *Nonlocal amplification of intense vorticity in turbulent flows*, Phys. Rev. Research **3** (2021), L042020.
- [3] J. Chen and T. Y. Hou, *Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data*, I: [arXiv:2210.07191](#); II: [arXiv:2305.05660](#).
- [4] P. Constantin and G. Iyer, *A stochastic Lagrangian representation of the three-dimensional incompressible Navier–Stokes equations*, Comm. Pure Appl. Math. **61** (2008), 330–345.
- [5] T. M. Elgindi, *Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$* , Ann. of Math. **194** (2021).
- [6] S. Goto, Y. Saito and G. Kawahara, *Hierarchy of antiparallel vortex tubes in spatially periodic turbulence at high Reynolds numbers*, Phys. Rev. Fluids **2** (2017), 064603.
- [7] B. E. Hamlington, J. Schumacher and W. J. A. Dahm, *Local and nonlocal strain rate fields and vorticity alignment in turbulent flows*, Phys. Rev. E **77** (2008), 026303.
- [8] A. Kiselev and V. Šverák, *Small scale creation for solutions of the incompressible two dimensional Euler equation*, Ann. of Math. **180** (2014), 1205–1220.
- [9] T. Mahithitarmmatorn, *Exact mean–covariance dynamics of the Weber field in the stochastic Lagrangian representation of the 3D Navier–Stokes equations*, [arXiv:2608.16915](#) (2026).
- [10] E. Miller, *Finite-time blowup for a Navier–Stokes model equation for the self-amplification of strain*, Anal. PDE **16** (2023).
- [11] J. Nečas, M. Růžička and V. Šverák, *On Leray's self-similar solutions of the Navier–Stokes equations*, Acta Math. **176** (1996), 283–294.
- [12] T.-P. Tsai, *On Leray's self-similar solutions of the Navier–Stokes equations satisfying local energy estimates*, Arch. Ration. Mech. Anal. **143** (1998), 29–51.

- [13] OpenAI, *Finite time blowup for the Euler equation*, preprint (2026), not independently verified at the time of writing.
- [14] T. Tao, *Finite time blowup for an averaged three-dimensional Navier–Stokes equation*, J. Amer. Math. Soc. **29** (2016), 601–674.
- [15] T. Tao, *Finite time blowup for Lagrangian modifications of the three-dimensional Euler equation*, Ann. PDE **2** (2016). [arXiv:1606.08481](#).
- [16] A. Salkimbayev, *Transport and the supercriticality gap in 3D Navier–Stokes, I*, preprint (2026).
- [17] A. Salkimbayev, *Transport and the supercriticality gap in 3D Navier–Stokes, II: The flow map*, preprint (2026).
- [18] A. Salkimbayev, *Transport and the supercriticality gap in 3D Navier–Stokes, IV: Events*, preprint (2026).

ALMATY, KAZAKHSTAN

Email address: `adilsalkimbayev@gmail.com`